① Define 4-connectivity, 8-connectivity and m-connectivity. ~~→6~~ with suitable example -6.

② Write notes on pixel relationship and distance measurement techniques for image processing with suitable ~~reta~~ example.

Ans. to ques. no: 01

Connectivity: Connectivity describes how two pixels are connected to each other. There are 3-types of connectivity:

i) 4-connectivity

ii) 8-connectivity

iii) m-connectivity.

i) 4-connectivity: Two pixels are 4-connected when they are 4-neighbours and have same intensity value.

Set of 4-neighbours:

$$N_4(p) = \{(x+1, y), (x-1, y), (x, y+1), (x, y-1)\}$$

Example:

84	54	8	1
20	5	7	2
110	50	80	20

$V = \{1, 2, 3, \ldots, 10\}$

4-connectivity

ii) 8-connectivity: Two pixels are 8-connected when they are 8-neighbours and have same intensity value. It also includes diagonal neighbours.

Example:

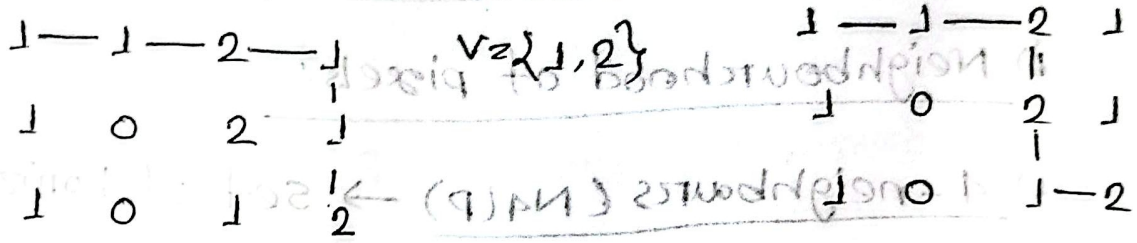
84	54	8	1
20	5	7	2
110	50	80	20

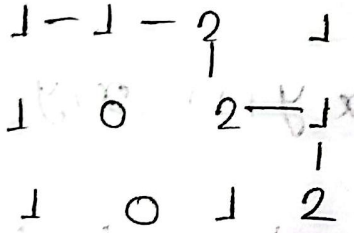
8-connectivity

$V = \{1, 2, 3, \ldots, 10\}$

iii) m-connectivity: m-connectivity removes ambiguity caused by diagonal connections in 8-connectivity.

Example:





Here numerous solution is available. But m-connectivity helps to choose the shortest path with creating ambiguity.

Ans. to the ques. no. 2

Pixels are the smallest element that form an image.

Relationship between pixels:

a) Neighbourchood of pixels:

i) 4-neighbours (N4(P)) → set of horizontal and vertical neighbours.

$N_4(P) \rightarrow \{(x+1, y), (x-1, y), (x, y+1), (x, y-1)\}$

ii) 8-neighbours (N8(P)): set of 4 neighbours and diagonal neighbours.

iii) Diagonal neighbours (ND(P)): set of Diagonal neighbours.

$N_D(P) = \{(x+1, y-1), (x+1, y+1), (x-1, y-1), (x-1, y+1)\}$

Distance measurement of pixels:

i) **Euclidean distance:** straight line distance on 2D plane between two pixels.

$$De(p,q) = |(x-s)^2 + (y-t)^2|^{1/2}$$

here $p(x,y)$, $q(s,t)$ are two pixels.

ii) **City-Block distance:**

$$D4(p,q) = |x-s| + |y-t|$$

Distance of horizontal and vertical pixels in image. Knows as mahhattan distance.

iii) **Chessboard distance:**

$$D8(p,q) = \max(|x-s|, |y-t|)$$

Distance between 8-neighbours. It includes 4-neighbours and diagonal neighbours. It is connected to 8-connectivity.

CamScanner